

Washington Cardoso Pereira — Full Stack Developer

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Professional Summary

Full Stack Engineer with 4 years of experience building production web applications, backend APIs and multi-tenant SaaS with Ruby on Rails, TypeScript and SQL. Focused on technical ownership, backend performance, resilient architecture and AI-assisted engineering, with measurable results ranging from queries running 63,000x faster to external integrations sustaining 99.9% uptime.

Professional Experience

Full Stack Developer | was Advisory

Aug 2023 - Present

- Optimized a critical endpoint, cutting latency from 3.5s to 180ms (-94%) by identifying N+1 queries, implementing eager loading and 3-layer Redis caching, impacting 10k+ requests/day.
- Optimized PostgreSQL queries on a base with 14M+ records, designing composite indexes and refining query strategies that cut response time from 12.7s to 0.2ms (~63,000x).
- Rewrote a critical PL/pgSQL function (V6) that was blowing statement_timeout on a tenant with 131k+ records, cutting execution time from 28.7s to 12.1s (~10x) via a materialized CTE and dedicated tuning of work_mem and JIT; validated with a dedicated performance spec.
- Fixed a critical data isolation vulnerability in a public tax exemption portal, where a swallowed exception (silent rescue) was masking a broken .includes, exposing 242k+ cross-tenant records and causing timeouts. Rebuilt the scope with UNION ALL and preload before pagination: latency 504ms to 70ms, and the cross-tenant leak was closed.
- Rearchitected the CSV report export pipeline, replacing synchronous processing with an asynchronous architecture using Sidekiq, PostgreSQL COPY, streaming and Amazon S3, reducing export time of 1.77M records from ~80h to 7min (~600x).
- Architected and implemented the integration with the external SINAL API, documented in OpenAPI/Swagger, processing 100k+ records/month in a multi-tenant environment, applying Faraday, a distributed Circuit Breaker (Stoplight), Retry with Exponential Backoff, the Outbox Pattern, PostgreSQL advisory locks and Sidekiq for asynchronous processing, publishing events to AWS EventBridge and Amazon SQS and deployed on Kubernetes, sustaining 99.9% uptime and zero duplicate records even with an unstable provider.
- Reduced write cost on high-volume tables by removing 33 unused indexes and adding unique indexes on join tables via concurrent migrations (algorithm: :concurrently) applied with zero downtime, and fixed a race condition emitting duplicate protocol numbers under double-click by replacing an unlocked SELECT MAX(number)+1 with pg_advisory_xact_lock scoped by department and year, with an explicit transaction guard and a uniqueness validation as second defense layer.
- Led the automated testing architecture for the Compras project: distributed tasks via Trello, reviewed deliveries and unblocked the team in real time, and used a custom Claude Code agent skill to accelerate onboarding, cutting production incidents by ~89% as monitored through Datadog observability.
- Participated in the Domain-Driven Design architecture of the Auditoria project alongside the tech lead, including introducing Devise as the authentication solution and shaping the bounded contexts of the module.
- Built automated test suites with RSpec and Capybara, raising coverage from 35% to 85% and reducing production bugs through robust CI/CD pipelines, with RuboCop and Brakeman for static analysis and security, shipped through a feature-branch Git workflow.
- Evolved the critical Acordos de Divida Ativa module in a multi-tenant MVC architecture serving 50+ municipalities, during the system migration from Rails 7 to Rails 8, structured with Devise, Service Objects and Form::Builder to encapsulate complex fiscal rules, delivering 15+ per-municipality customizations via feature flags, cutting delivery time by 60%, zeroing cross-tenant regressions and reducing module-related support tickets by 59%.
- Optimized production MySQL queries alongside the PostgreSQL work, tuning schema and index design for the access patterns that mattered.
- Monitored production applications using observability tooling (Datadog), identifying and resolving incidents proactively before they affected end users.
- Built and continuously evolve Swarm, an Agent Development Environment on Rails 8 and Hotwire that orchestrates up to 4 parallel Claude Code agents per prompt, each in an isolated Git worktree with a real-time PTY over ActionCable, a compare grid to pick the best attempt, live diff against the merge-base with per-line comments that flow back into the agent session, one-click --no-ff merge or PR creation via the GitHub CLI, an embedded preview and a CodeMirror 6 editor; shipped as a native desktop app with Tauri 2 and WebView2 on Windows, Rails backend inside WSL, i18n in pt-BR and English.
- Published a trilogy of Claude Code plugins covering agent input, output and side effects: lean-output compresses RSpec, RuboCop, Brakeman, git diff/show and grep output by -29% to -95% tokens (-49% overall, measured against a real corpus of 251 transcripts, 393kB to 202kB, not synthetic benchmarks) with a zero-loss guarantee on failures; rails-context injects a curated dossier of columns, indexes, associations, validations and routes instead of loading full db/schema.rb and models; rails-guard intercepts destructive Rails commands via a PreToolUse hook, requiring human confirmation before execution.
- Worked in a Dockerised environment at Nobe Sistemas, adjusting the Docker and nginx configuration of the Tributario service within an existing container setup.
- Built ObraHub, the labor tracking and billing system in production at Engemil (industrial maintenance and assembly): a Rails 8 + PostgreSQL + Hotwire monolith that replaced a paper-to-spreadsheet-to-retyping cycle, with idempotent SHA-256 row imports and a monthly billing workbook exported byte-identical to the client's reference file; in real use, 11,551h closed in one month and 1,395 entries imported and validated in 3s.
- Built shelflog, an installable Rails PWA for shelf expiry-date control, designed for use in Muffato's PAS (a major Brazilian supermarket chain): barcode and name lookup, a per-product status driven by configurable thresholds, and an hourly Solid Queue job that sends web push alerts for expired and soon-to-expire stock.

Full Stack Developer GoTech DS	Jun 2023 - Jul 2023
Contributed to feature implementation on the Go Tech DS product using Ruby on Rails and Devise, developing the authentication flow (login, signup and password recovery) and dashboard, following Trello cards within the existing architecture, and participating actively in code reviews and pair programming.	
Full Stack Developer Freelance & Trainee	Nov 2022 - May 2023
Delivered blogs, landing pages and mini-softwares end to end for direct clients as the single owner of every project: requirements, PostgreSQL and MySQL modeling, Ruby on Rails backend, frontend, deploy and ongoing support, setting up CI/CD pipelines, domain configuration, SSL and monitoring for each handoff, and handling client communication, scope negotiation and pricing; picked up Stimulus, Tailwind, AWS S3 and payment gateways on demand.	

Projects & Open Source

- Built PipelineHQ, a full-stack multi-tenant B2B CRM on Rails 8.1 with a Pipedrive-style sales pipeline, using Solid Queue, Solid Cache and Solid Cable (no Redis, no Sidekiq), Hotwire with Turbo and Stimulus, Tailwind v4, and native Rails authentication extended with 10 hardening layers including TOTP 2FA, backup codes, sudo mode, rate limiting, audit log and honeypot, with every decision backed by an ADR.
- Published a Ruby wrapper for BrasilAPI as a first dive into open source, handling public code, gem packaging and community feedback.
- Developed Tally for a client, a financial reconciliation platform now entering production, with a Kotlin and Spring Boot engine, PostgreSQL persistence and a Rails console; designed streaming ingestion for 100k-line bank files at 9,405 lines/s end to end, cent-accurate ledger conservation, human review workflows, REST/OpenAPI contracts and 199 automated tests across domain, integration and UI layers.
- Built a two-service telehealth platform with a Rails 8.1 GraphQL clinical API and a Django REST billing service, each owning its PostgreSQL database; implemented encrypted medical records, role-based access and audit trails alongside idempotent Stripe webhooks, asynchronous Sidekiq and Celery workflows, Redis, API security, service-to-service contracts and containerized deployments.
- Built Resumo de Ocorrencias, a Python and Tkinter desktop application that consolidates large Excel production reports, automatically detects changing workbook layouts, filters operational incidents and visualizes production metrics; redesigned XLSX ingestion with openpyxl streaming and direct worksheet XML parsing, loading a real 527MB, 8-workbook package in 19 seconds while keeping the UI responsive and producing results identical to the full reader.
- Building Glossa, a translation management platform with a Rails 8.1 JSON:API backend designed for an Ember.js client, backed by PostgreSQL and developed through strict test-first cycles; implemented YAML and JSON locale round-tripping that preserves untouched bytes, source-change detection, placeholder and plural validation, review workflows, role- and locale-scoped authorization, background imports and stable machine-readable API errors.

Technical Skills

- Backend & Languages:** C#, Elixir, JavaScript, Kotlin, Python, Ruby, Rust, SQL, Shell Script, TypeScript, .NET, Django, Electron, Ember.js, Hotwire, Next.js, Node.js, Phoenix, React, React Native, Ruby on Rails, Sinatra, Spring Boot, Stimulus, Tailwind CSS, Tauri
- Databases & Infra:** Elasticsearch, MySQL, PostgreSQL, Redis, SQL Server, SQLite, AWS, ArgoCD, CI/CD, Datadog, Docker, Git/GitHub, Grafana, Kubernetes, Linux, OpenTelemetry, RabbitMQ, Sidekiq, Solid Queue
- Testing & Practices:** Brakeman, Capybara, RSpec, RuboCop, Selenium, TDD, Testcontainers, Claude API, Clean Architecture, Event-Driven, GraphQL, MCP, Multi-tenant SaaS, RAG, RESTful APIs, WebSockets

Languages

English: B2 (technical reading, writing and speaking)
Portuguese: native

Educational Background

Systems Analysis and Development (ADS) — in progress, expected Dec 2026 Wyden — Sao Jose do Rio Preto, SP	2024 - 2026
Treina-Dev — Campus Code, Group 10	2023
AWS Serverless Demonstrated — issued Sep 2026, expires Sep 2027 Amazon Web Services (AWS)	2026